



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Sequence in Metric Space

Cauchy Sequence and Completeness Theorems

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CONDUCTED BY

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Sequence in Metric Space

Sequence in Metric Space

Let (X, d) be a metric space. A sequence in X is a function $f : \mathbb{N} \rightarrow X$. We usually denote the sequence by $\{x_n\}$, where $x_n = f(n)$ for each $n \in \mathbb{N}$.

Convergence of a Sequence in $\mathbb{R}$

Convergence of a Sequence

A sequence $\{a_n\}$ is said to **converge** to a real number a if for every real number $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $n \geq N$, the terms of the sequence satisfy the inequality

$$|a_n - a| < \varepsilon.$$

If such a number a exists, it is called the **limit** of the sequence, and we write

$$\lim a_n = a \quad \text{or} \quad a_n \rightarrow a \text{ as } n \rightarrow \infty.$$

Convergence of a Sequence in Metric Space

Convergence of a Sequence in Metric Space

Let (X, d) be a metric space. A sequence $\{x_n\}$ in X is said to **converge** to a point $x \in X$ if for every $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $n \geq N$, the terms of the sequence satisfy the inequality

$$d(x_n, x) < \varepsilon.$$

If such a point x exists, it is called the **limit** of the sequence, and we write

$$\lim x_n = x \quad \text{or} \quad x_n \rightarrow x \text{ as } n \rightarrow \infty.$$

Cauchy Sequence in Metric Space

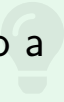
Cauchy Sequence

Let (X, d) be a metric space. A sequence $\{x_n\}$ in X is called a **Cauchy sequence** if for every $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $m, n \geq N$, the terms of the sequence satisfy the inequality

$$d(x_n, x_m) < \varepsilon.$$

Every Convergent Sequence is Cauchy

Let (X, d) be a metric space. If a sequence $\{x_n\}$ in X converges to a point $x \in X$, then $\{x_n\}$ is a Cauchy sequence.



💡 Not Every Cauchy Sequence is Convergent

Let (X, d) be a metric space. It is not necessarily true that every Cauchy sequence in X converges to a point in X .

Examples of Sequences in Metric Spaces

❓ Problem

Let

$$X = \{0\} \cup \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}.$$

Define two metrics on X by

$$d_1(x, y) = |x - y|, \quad d_2(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Consider the sequence

$$x_n = \frac{1}{n}.$$

Show that (x_n) converges in the metric space (X, d_1) but does not converge in the metric space (X, d_2) .

❓ Problem

Let $X = C[0, 1]$, the space of all continuous real-valued functions on $[0, 1]$. Define two metrics on X by

$$d_1(f, g) = \int_0^1 |f(x) - g(x)| dx, \quad d_2(f, g) = \sup_{x \in [0, 1]} |f(x) - g(x)|.$$

Consider the sequence of functions

$$f_n(x) = x^n.$$

Show that (f_n) converges in the metric space (X, d_1) but does not converge in the metric space (X, d_2) .

Completeness of a Metric Space

Complete Metric Space

A metric space (X, d) is called **complete** if every Cauchy sequence in X converges to a point in X .

Completeness of $\mathbb{R}^n$

$\mathbb{R}$ is a Complete Metric Space

The set of real numbers $\mathbb{R}$ with the usual metric $d(x, y) = |x - y|$ is a complete metric space. 

$\mathbb{Q}$ is not a Complete Metric Space

The set of rational numbers $\mathbb{Q}$ with the usual metric $d(x, y) = |x - y|$ is not a complete metric space. 

$\mathbb{R}^n$ is a Complete Metric Space

The set of n -tuples of real numbers $\mathbb{R}^n$ with the usual metric 

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

is a complete metric space.

Completeness of ℓ^p Spaces

💡 ℓ^p is a Complete Metric Space

The set of sequences ℓ^p with the metric

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^{\infty} |x_i - y_i|^p \right)^{1/p} \quad \mathbf{x}, \mathbf{y} \in \ell^p$$

is a complete metric space.



💡 ℓ^∞ is a Complete Metric Space

The set of sequences ℓ^∞ with the metric

$$d(\mathbf{x}, \mathbf{y}) = \sup_{i \in \mathbb{N}} |x_i - y_i| \quad \mathbf{x}, \mathbf{y} \in \ell^\infty$$

is a complete metric space.



Completeness of Function Spaces

❓ $C[0, 1]$ under sup metric is a Complete Metric Space

Let $C[0, 1]$ be the set of all continuous real-valued functions defined on the interval $[0, 1]$. Define a metric d on $C[0, 1]$ by

$$d(f, g) = \sup_{x \in [0, 1]} |f(x) - g(x)|$$

for all $f, g \in C[0, 1]$. Show that $(C[0, 1], d)$ is a complete metric space.

❓ $C[0, 1]$ under integral metric is not Complete

Let $C[0, 1]$ be the set of all continuous real-valued functions defined on the interval $[0, 1]$. Define a metric d on $C[0, 1]$ by

$$d(f, g) = \int_0^1 |f(x) - g(x)| dx$$

for all $f, g \in C[0, 1]$. Show that $(C[0, 1], d)$ is not a complete metric space.

Completeness and Closed Sets

💡 Closure of a Set and Convergence of Sequences

Let (X, d) be a metric space and let A be a subset of X . A point $x \in X$ is in the closure of A if and only if there exists a sequence $\{a_n\}$ in A that converges to x .

Closed subspace of a Complete Space is Complete

Let (X, d) be a complete metric space and let (Y, d) be a subspace of X . Then the metric space (Y, d) is complete if and only if Y is closed in X .

Cantor's Intersection Theorem

💡 Cantor's Intersection Theorem

Let (X, d) be a complete metric space. If $\{F_n\}$ is a sequence of non-empty closed subsets of X such that $F_{n+1} \subseteq F_n$ for all $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} \text{diam}(F_n) = 0$, then the intersection $\bigcap_{n=1}^{\infty} F_n$ contains exactly one point.

Thank You!

We'd love your questions and feedback.

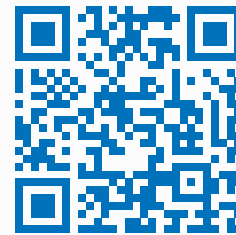
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(Lectures, walkthroughs, and course updates)



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References

- [1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.
- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.